STUDENT SERVICES / UC-STU-02

Student-Initiated Room Allocation

A student requests a suitable university room for any permitted activity; the system selects the best eligible option and the responsible faculty administrator approves or rejects it.

Primary actor	Authenticated student organizer
Supporting actors	Designated administrator of the room-owning faculty; optional invited professor(s); C++ scheduling engine
Preconditions	Rooms have an owning faculty, capacity, facilities and eligibility rules; that faculty has a designated approver; dated calendar data is configured.

Main success scenario

1. Student enters purpose/category (study, student-led lecture, club activity or other), description, requested date/time and duration, total attendees, required facilities, and any optional faculty/building preference or professor invitees.
2. The system checks room eligibility, capacity, facilities, dated/valid recurring commitments and closures; a C++ strategy ranks feasible single-room candidates and explains its best match.
3. Student reviews the recommended room (or a feasible alternative) and submits. The request becomes Pending Faculty Approval; no room is held.
4. The request goes to the designated administrator of the selected room's owning faculty. Invited professors receive optional attendance invitations and may accept or decline independently.
5. The administrator approves or rejects. Before approval becomes confirmation, the system atomically rechecks current room availability and policy, then creates an exclusive reservation and records Confirmed.
6. The organizer sees status and invitation responses. Organizer or authorized room-faculty administrator may cancel; the reservation is released and changes are recorded and shown to relevant users.

Business rules and constraints

Automatic allocation	The student specifies requirements, not a mandatory exact room. Eligible rooms may be searched across faculties; ownership determines the approver. Room access restrictions still apply.
Selection strategy	Use suitable existing C++ greedy/best-fit concepts for one room, with deterministic ranking; do not misuse multi-room/shared-exam strategies. The ordinary student need not choose an algorithm.
Approval and conflicts	Pending requests never reserve. Only an approved, freshly revalidated candidate is reserved. Prevent conflicts with confirmed bookings and applicable university schedules, without breaking shared-exam rules.
Invitations and privacy	Professor invitations are optional and non-vetoing. Request purpose, invitees and response details are limited to authorized participants and faculty approvers; other users see permitted availability only.
Revision and clubs	Changing room/time/requirements requires a new approval. Official club affiliation must not be presented as verified without evidence; no separate club-management workflow is required.

Alternative outcomes

No match: explain unavailable rooms or constraints and offer feasible alternative rooms/times. Rejected: record rejection; no reservation. Stale recommendation: require a new candidate/revision and faculty review; never silently change the approved room. No authorized approver: block confirmation and explain the configuration issue.

Postconditions

Confirmed: one eligible room is exclusively reserved for the approved dated interval; relevant users are notified and history is retained. Rejected/cancelled/pending: no active room reservation remains.